Technical talk

Math and Diagrams

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Roadmap

1. Formalize the decision problem
2. Explain system structure
3. Connect results to evidence

Model

Sequential decisions under uncertainty

We consider an agent that repeatedly observes a state, chooses an action, and receives a reward.

- **State:** $s \in \mathcal{S}$ summarizes the information available now.
- **Action:** $a \in \mathcal{A}$ changes both reward and the next-state distribution.
- **Objective:** maximize expected discounted return over an indefinite horizon.

Modeling assumption

The current state is sufficient: conditional on s_t and a_t, the future is independent of the earlier history.

Bellman optimality equation

$$V^{\star(s)} = \max_a \quad R(s,a) \quad + \quad \gamma \quad \sum_{s'} P(s' \mid s,a) V^{\star(s')}$$

The recursion separates immediate utility from the expected value of all subsequent decisions [1].

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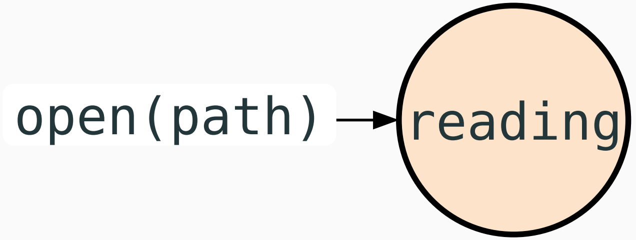
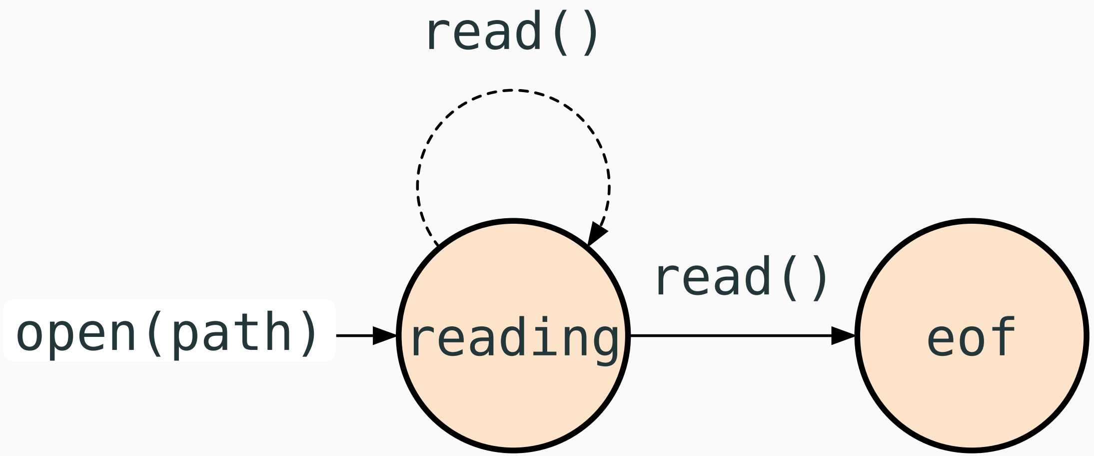
Structure

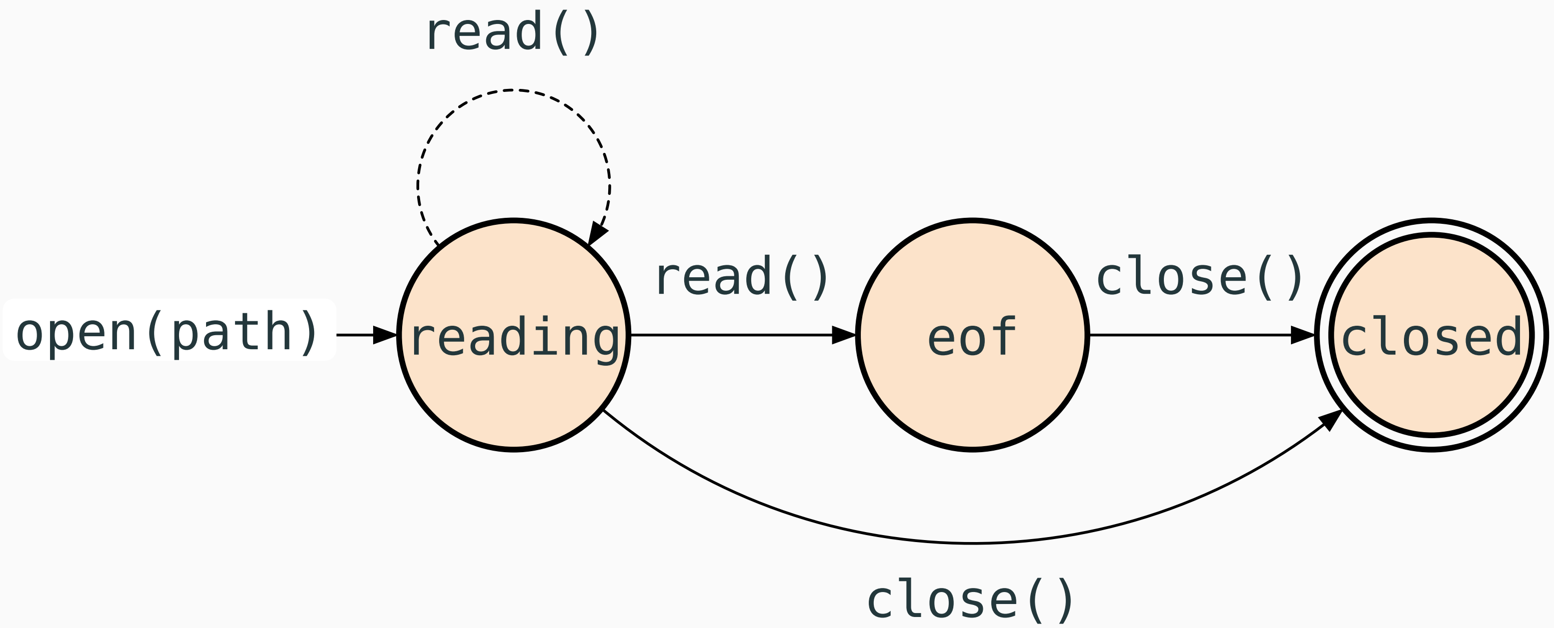
File-reader state machine



File-reader state machine

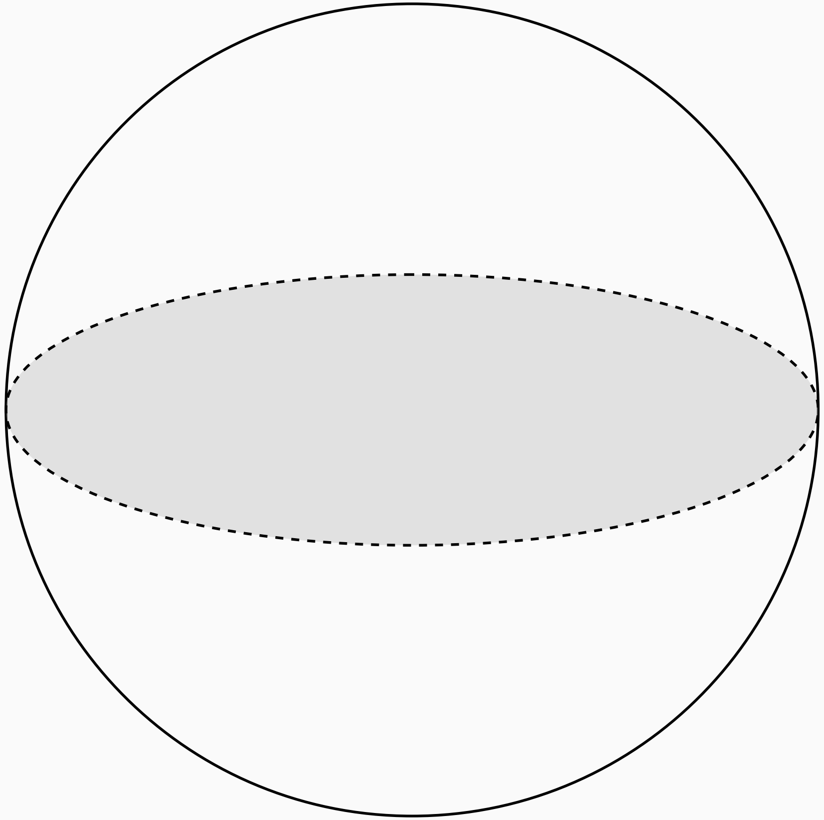


File-reader state machine

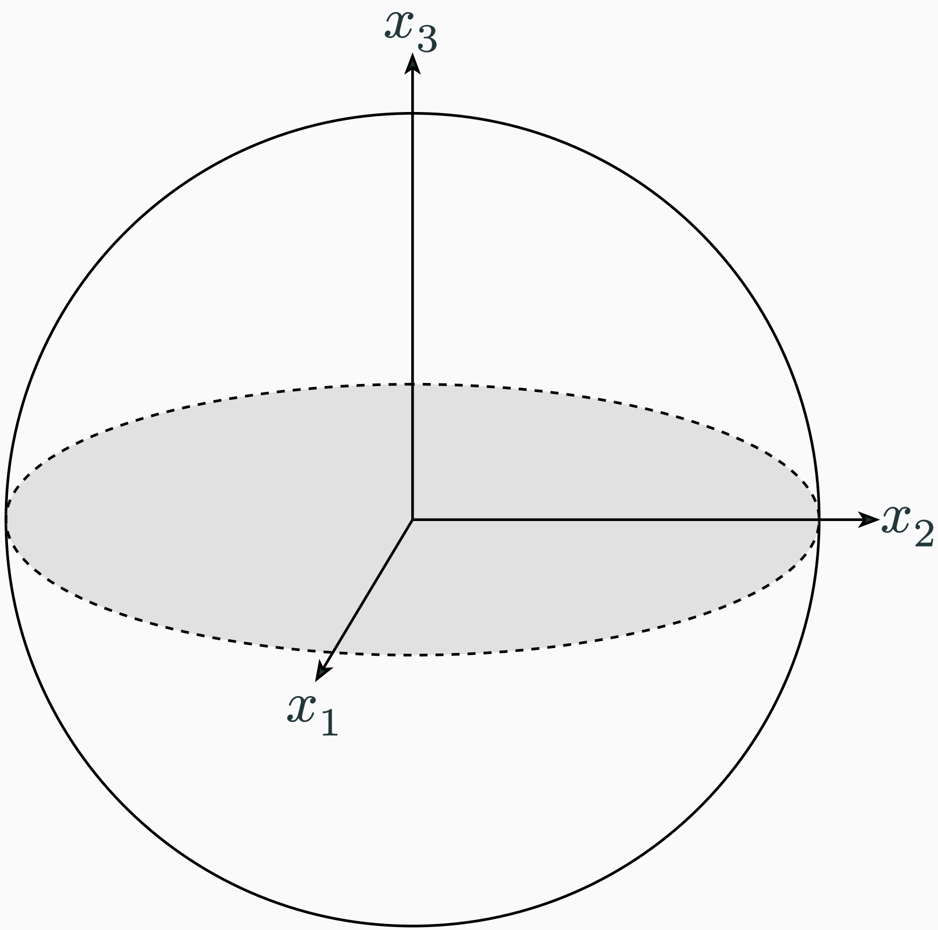


Geometry

A qubit as a Bloch vector



A qubit as a Bloch vector



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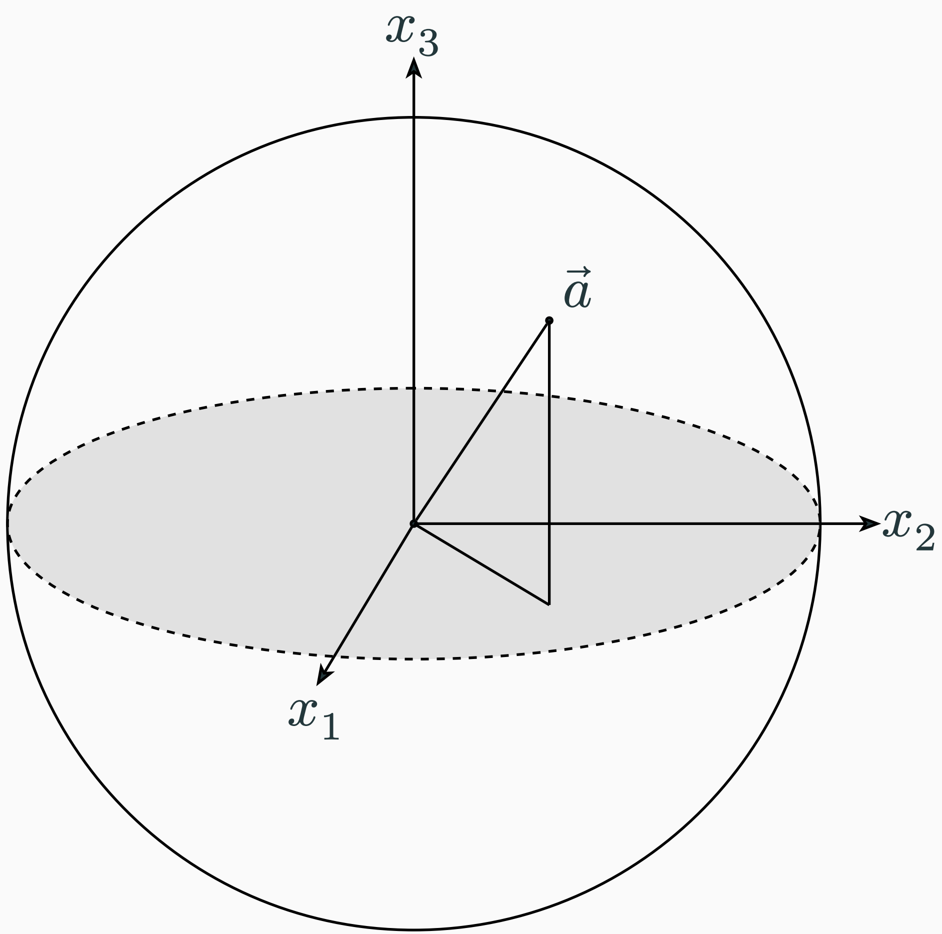

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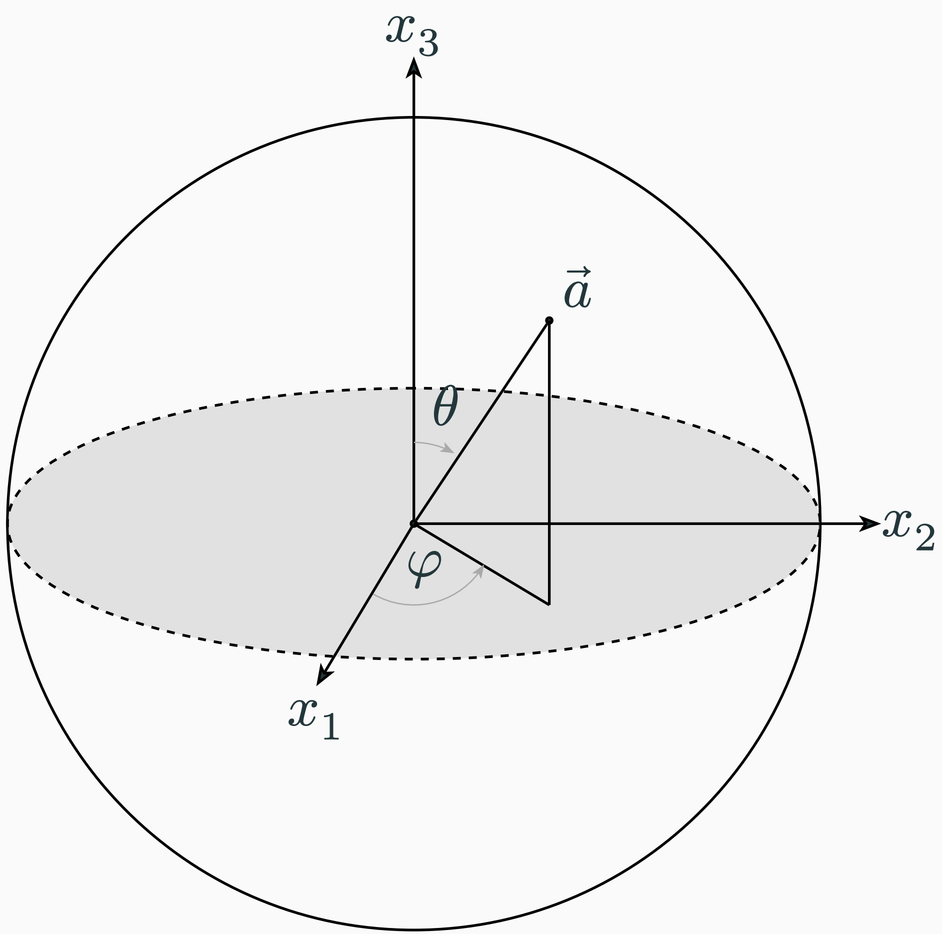

Evidence

What makes a technical slide?

Include

- a question or claim
- units and definitions
- reproducible computation
- a stated interpretation

Avoid

- unlabelled decorative plots
- numbers copied by hand
- animation without exposition
- citations disconnected from claims

Questions?

References

[1] R. Bellman, *Dynamic Programming*. Princeton University Press, 1957.